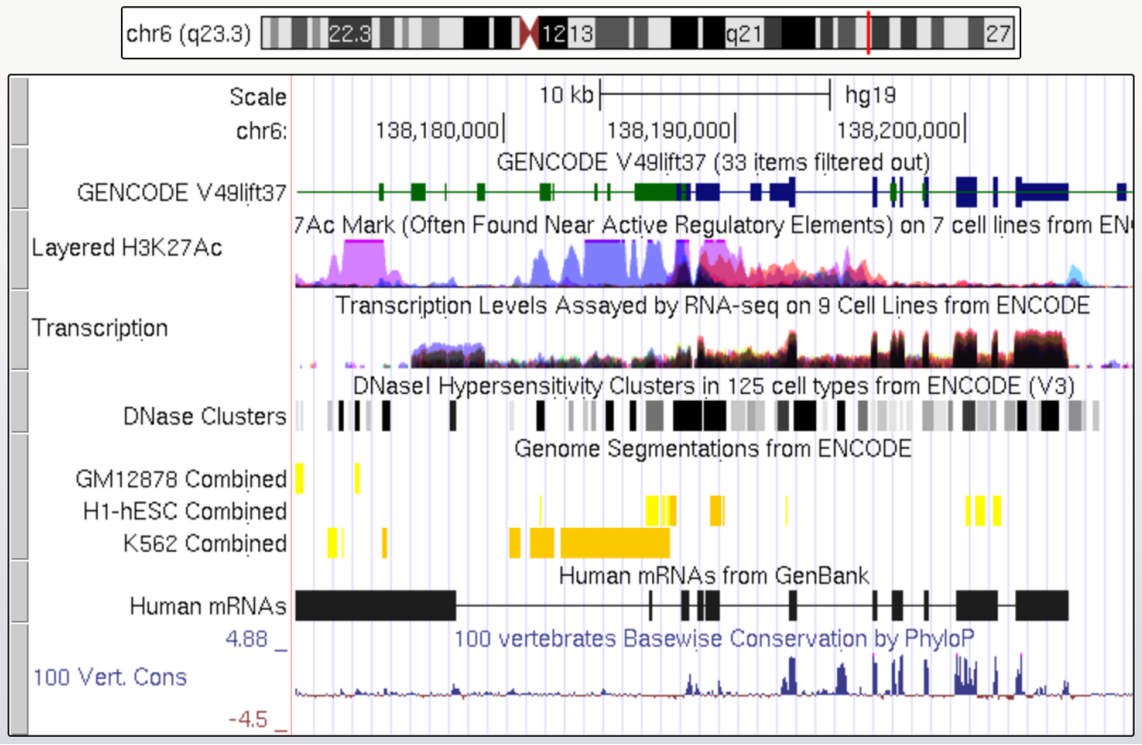
Sequence data are everywhere in biomedicine

genome data



proteins

Sequence[i]

Sequence status[i] Complete

Sequence processing[i] The displayed sequence is further processed into a mature form.

See also sequence in **UniParc** or sequence clusters in **UniRef**

Tools ▾ Download Add Highlight ▾ Copy sequence

Length 120
Mass (Da) 12,967
Last updated 2000-05-01 v1

MD5 Checksum[i] 5B52D672F6FF5F70FB93C 9E8BCCBFB38

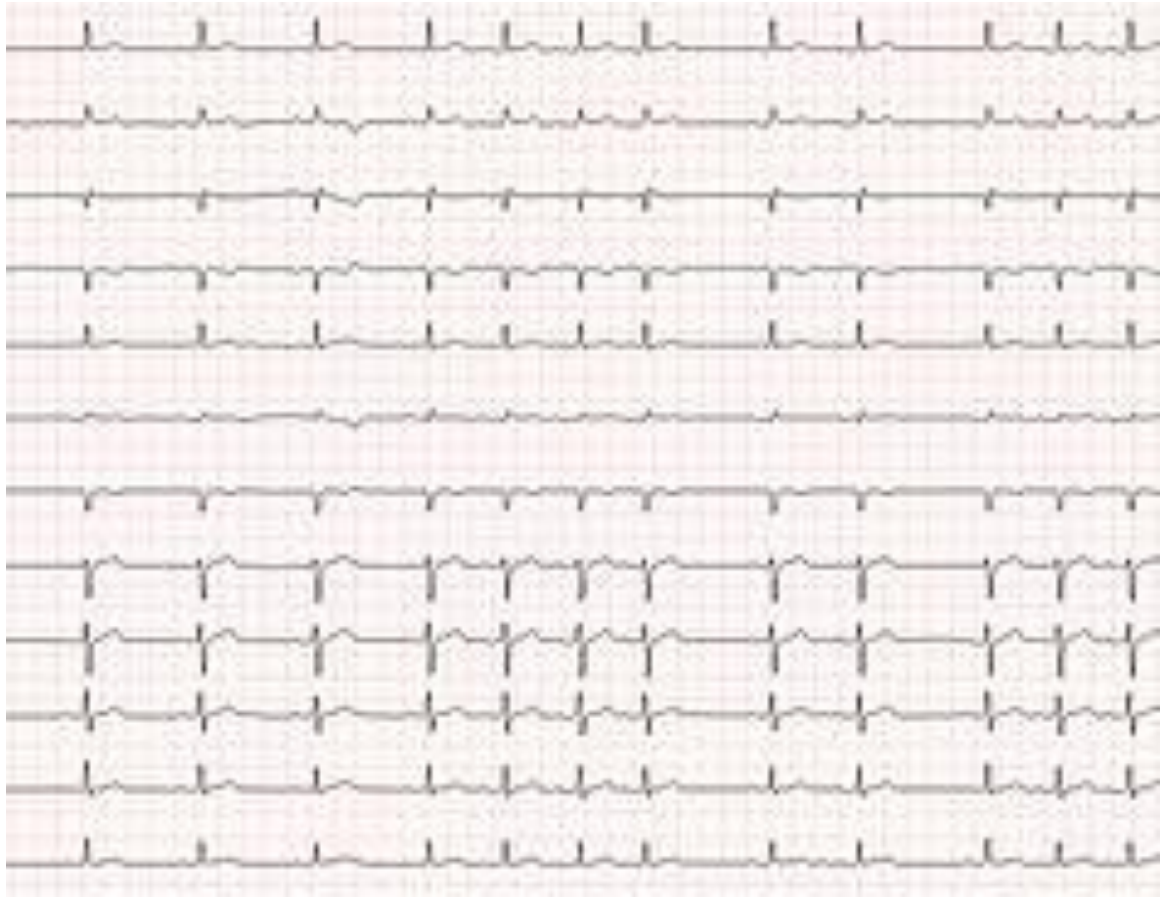
```
        10         20         30         40         50         60         70
MDKSFTLFLT LTILVVFIIS SPPVQAGFAN DLGGVAWATT GDNGSGCHGS IAECIGAEEE EMDSEINRRI

        80         90        100        110        120
LATTKYISYQ SLKRNSVPCS RRGASYYNCQ NGAQANPYSR GCSKIARCRS
```

genome.ucsc.edu, www.uniprot.org,

Sequence data are everywhere in biomedicine

electrocardiogram



research articles

EXPERIMENTAL PROCEDURES

Training and Evaluation of New MS^{2}PIP Spectrum Prediction Models for Immunopeptides

To train and test new MS^{2}PIP models, five publicly available immunopeptide data sets and one publicly available chymotrypsin-digestion data set were downloaded from PRIDE Archive (29, 30). Similarly, for evaluation on representative unseen data, four distinct data sets were downloaded: (i) a data set containing HLA-I immunopeptides, (ii) a data set containing HLA-II immunopeptides, (iii) the data set with tryptic peptides that was previously used to evaluate the existing MS^{2}PIP higher-energy collision-induced dissociation (HCD) models, and (iv) a data set containing chymotrypsin-digested peptide data. The corresponding ProteomeXchange project identifiers as well as the number of unique peptides and HLA patterns for each data set are listed in supplemental Table S1.

As tandem mass spectrometry (MS2) fragmentation patterns are highly dependent on the instrument, instrument settings, fragmentation method, and any applied labeling methods (25), all MS^{2}PIP train, test, and evaluation data must originate from experiments with the same experimental parameters. Unlabeled HCD data from

Figures: commons.wikimedia.org, 10.1016/j.mcpro.2022.100266

Symbols must first be converted into numbers

Manual feature engineering

ACDTK



	A	C	D	T	K
Basi	37	35	59	129	191
Heli	68	23	33	29	32
Hydr	51	75	25	35	0
pl	32	23	0	27	69

one-hot encoding

ACDTK



	A	C	D	T	K
A?	1	0	0	0	0
C?	0	1	0	0	0
D?	0	0	1	0	0
…	…	…	…	…	…
T?	0	0	0	1	0
…	…	…	…	…	…
K?	0	0	0	0	1
…	…	…	…	…	…

learned embedding

ACDTK



A	C	D	T	K
0.73	0.14	0.88	0.46	0.21
0.57	0.95	0.32	0.69	0.04
0.81	0.27	0.53	0.10	0.76
0.38	0.62	0.17	0.84	0.49
0.92	0.35	0.58	0.07	0.64
0.11	0.79	0.43	0.96	0.28
0.67	0.24	0.85	0.51	0.13
0.40	0.73	0.19	0.60	0.88

In sequence data, order matters

Cage the swallow.

≠

Swallow the cage.

```
MDKSFTLFLTLTILVVFIISSPPVQAGFANDLGGVAWATT
GDNGSGCHGSIAECIGAEEEEMDSEINRRILATTKYISYQ
SLKRNSVPCSRRGASYYNCQNGAQANPYSRGCSKIARCRS
```

≠

```
MDKSFTLFLTLTILVVFIISSPPVQAGFANDLGGVAWATT
GDNGSGCHGSIAECIGAEEDMEESEINRRILATTKYISYQ
SLKRNSVPCSRRGASYYNCQNGAQANPYSRGCSKIARCRS
```

In sequence data, memory matters as well

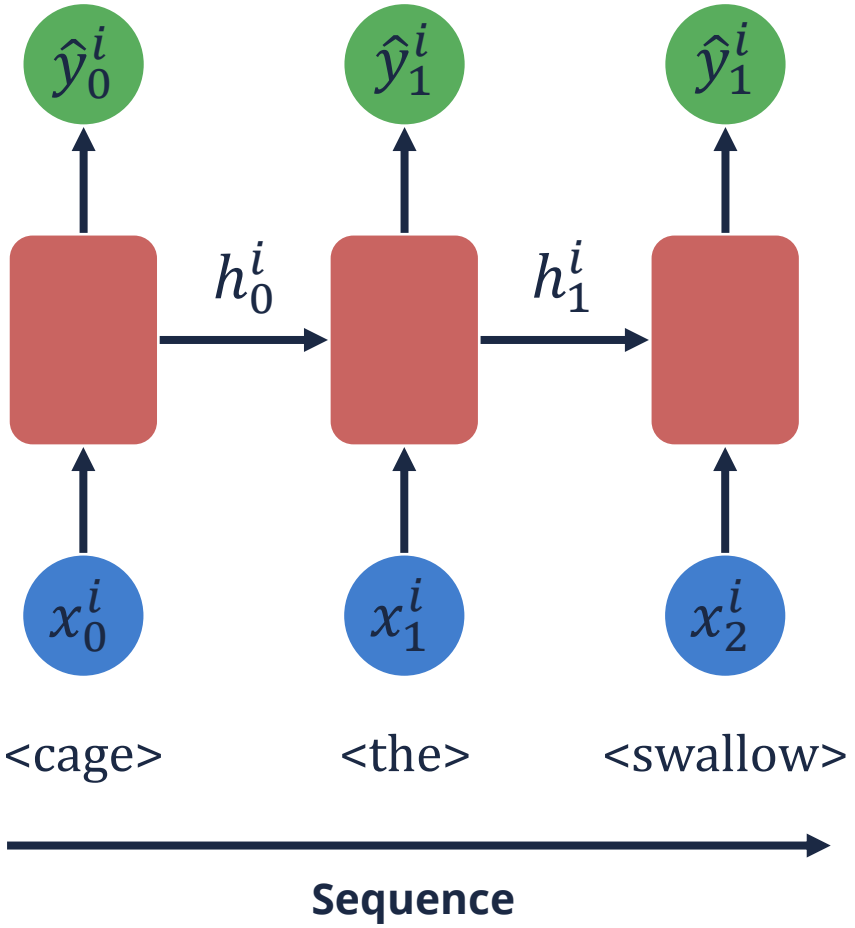
culture
office culture
cell culture

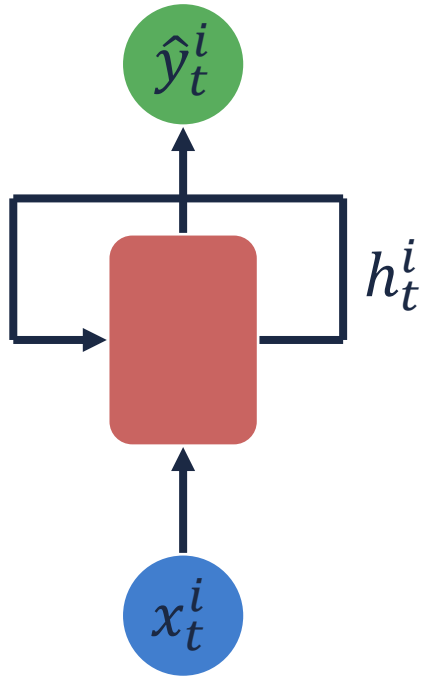
bank
river bank
investment bank

expression
artistic expression
gene expression

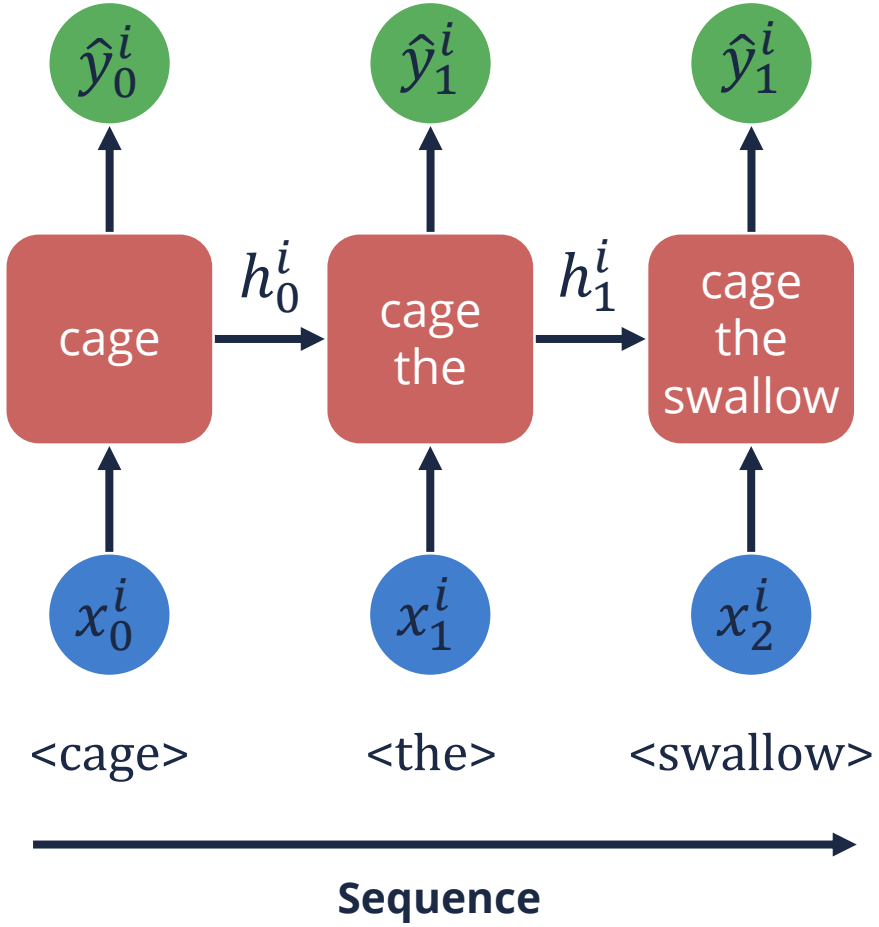
cell
immune cell
prison cell

Recurrent neural networks process a sequence one step at a time



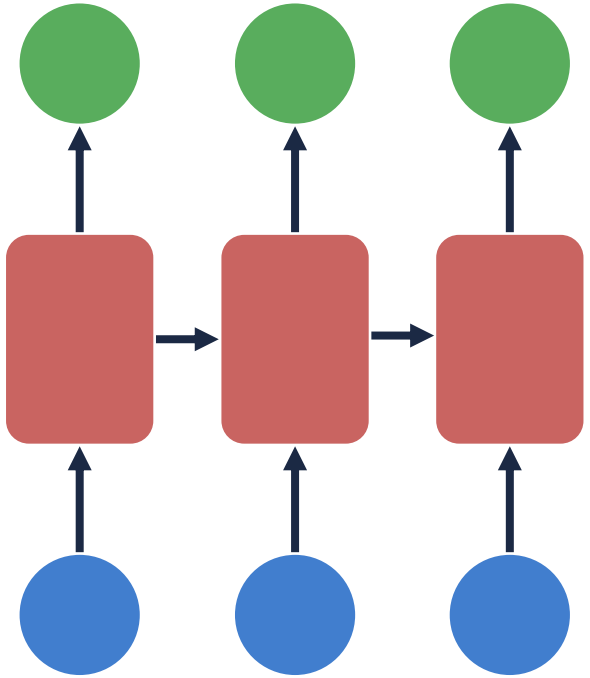


The hidden state acts as a running summary of the past

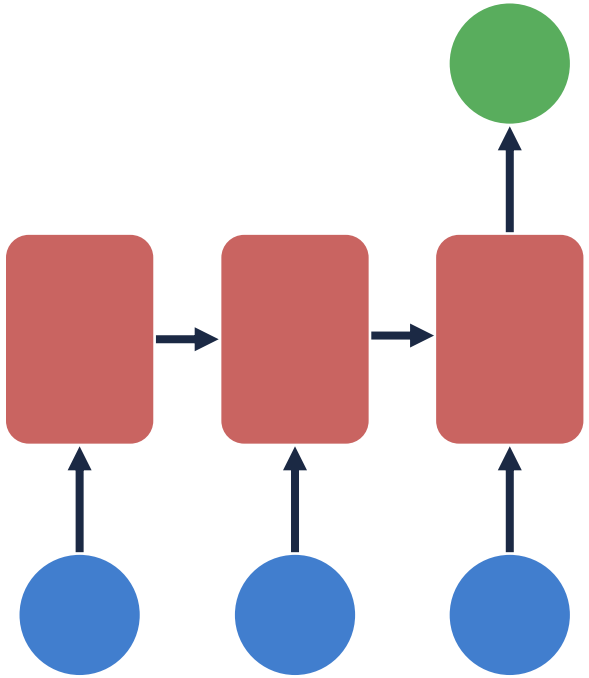


Recurrent neural networks can predict at each step or at the end

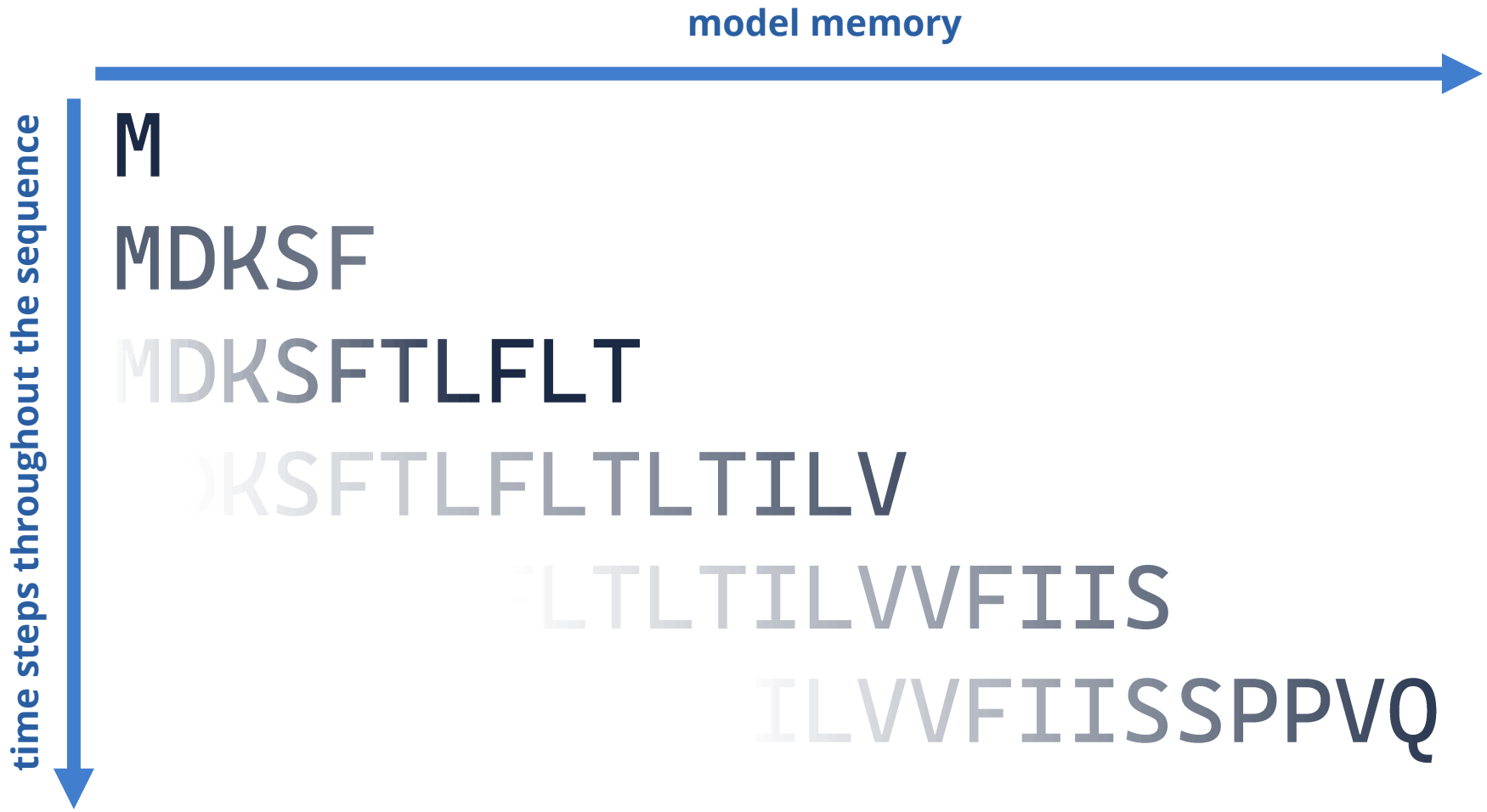
sequence to sequence
(many to many)



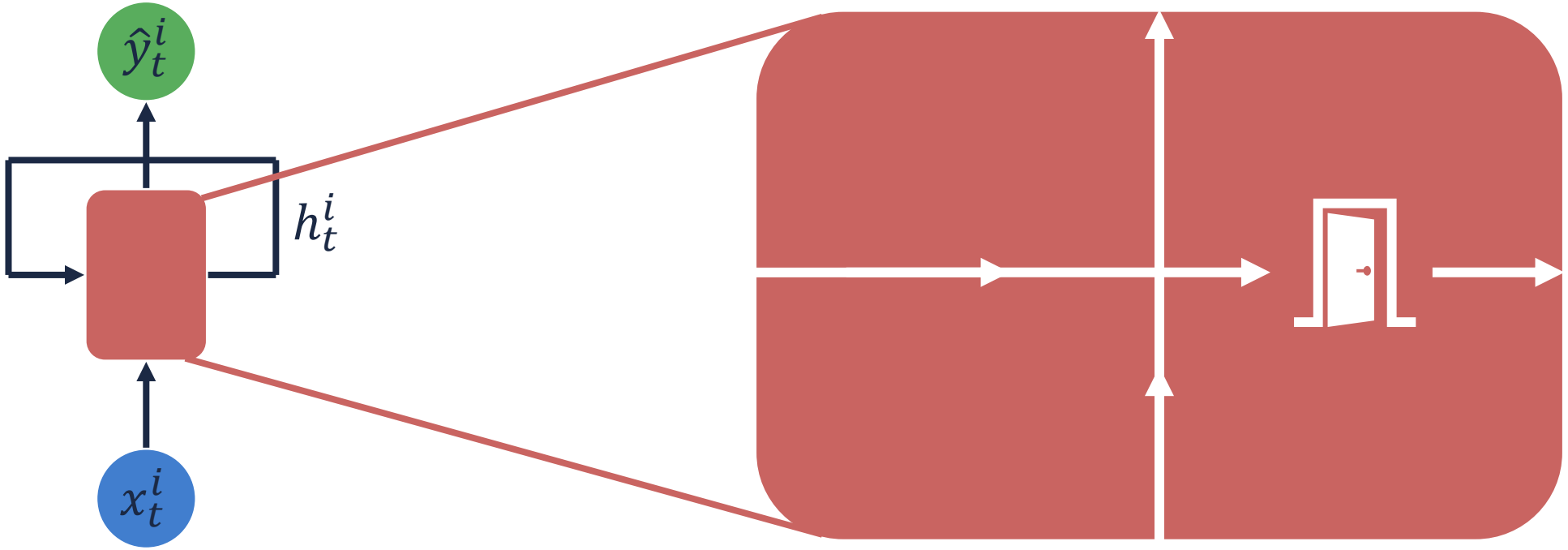
sequence to one
(many to one)



Vanilla RNNs struggle with long-range dependencies



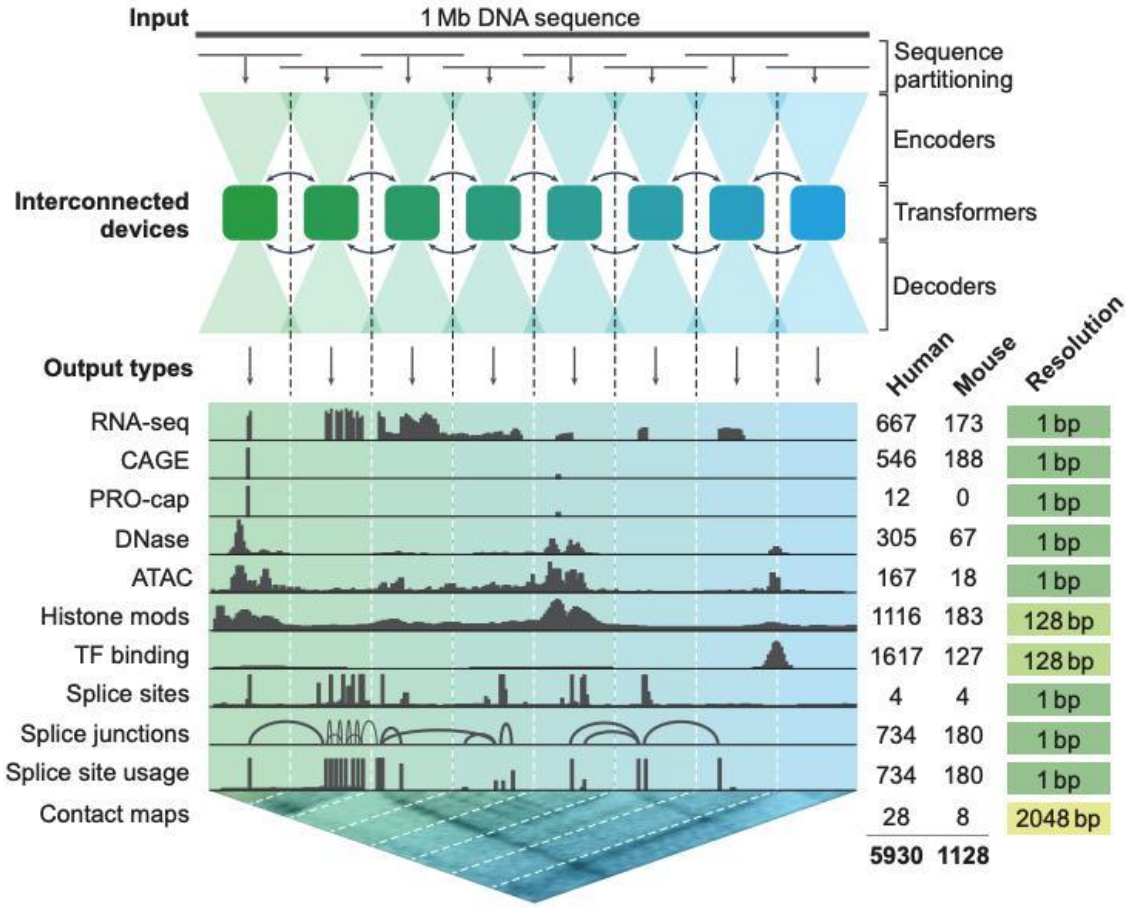
LSTMs and GRUs introduced gated memory to better preserve the important information



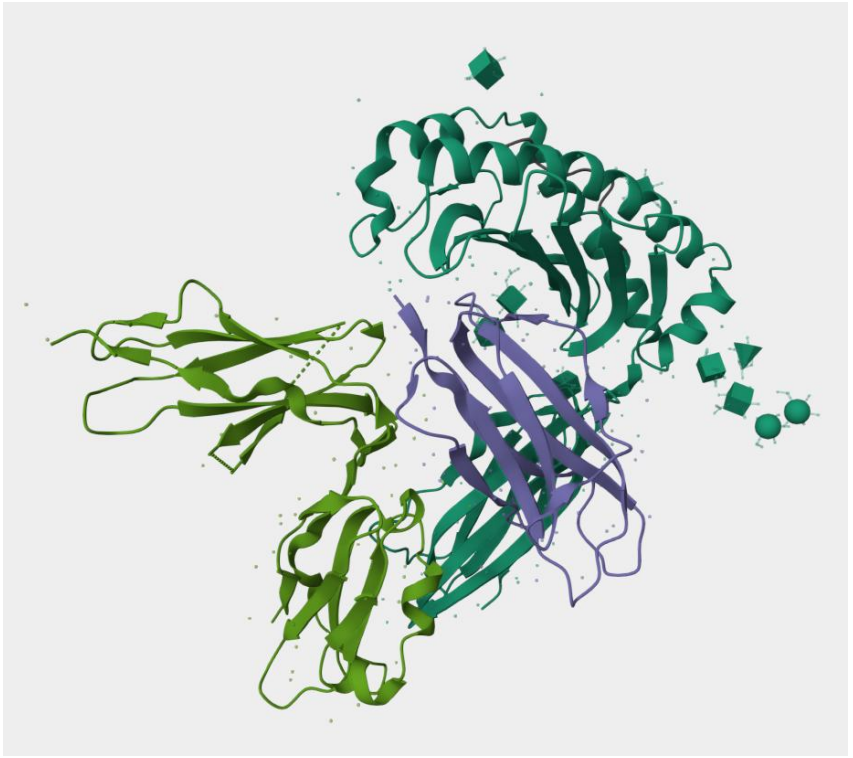
For complex tasks, recurrent memory is often not enough...

Especially when important context is far away

Long-range genome interactions

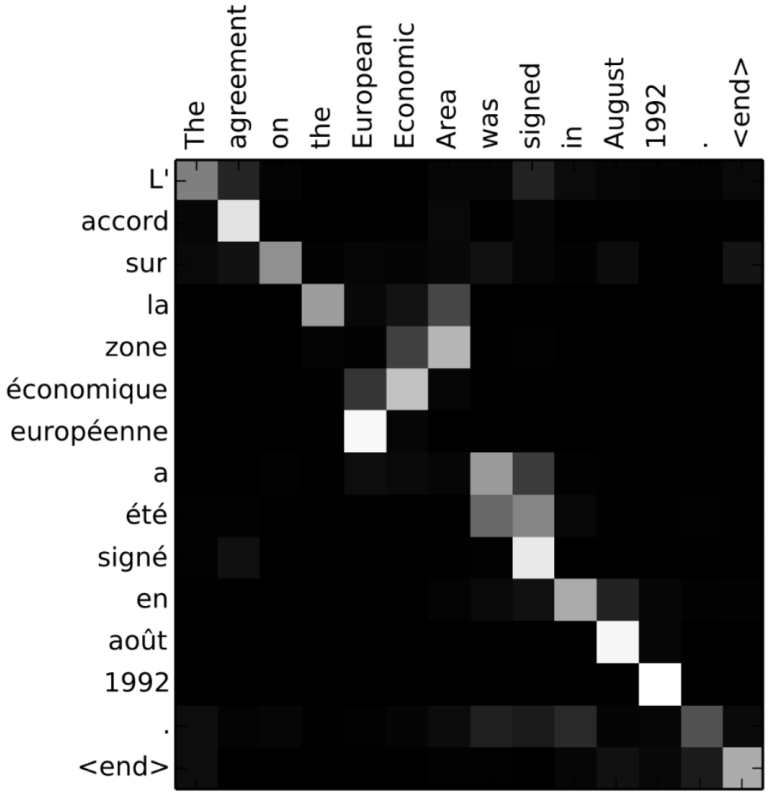


Amino acid interactions in 3D structure



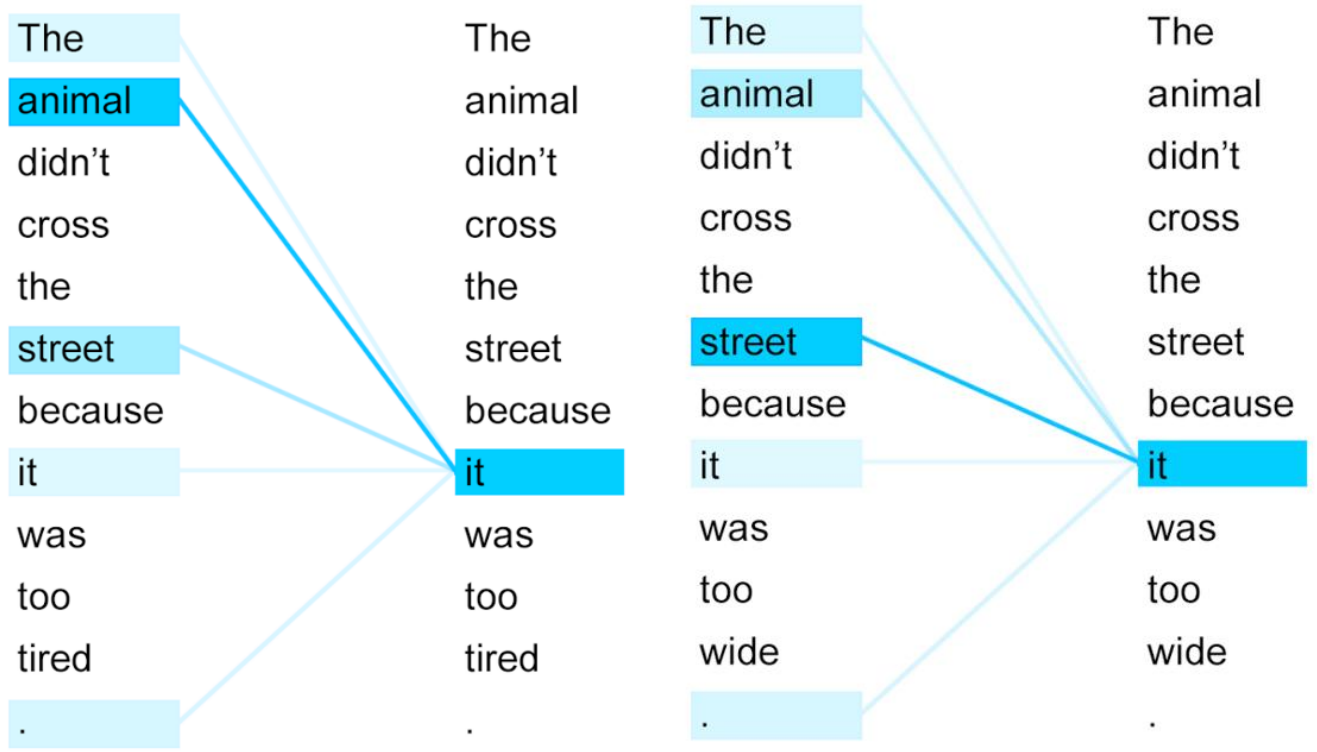
nature.com/articles/s41586-025-10014-0,uniprot.org/uniprotkb/P60709

Attention lets the model focus on the most relevant parts of the entire sequence



Attention-matrix heatmap

Bahdanau, et al. 2015. Neural machine translation by jointly learning to align and translate. In Proc. ICLR.

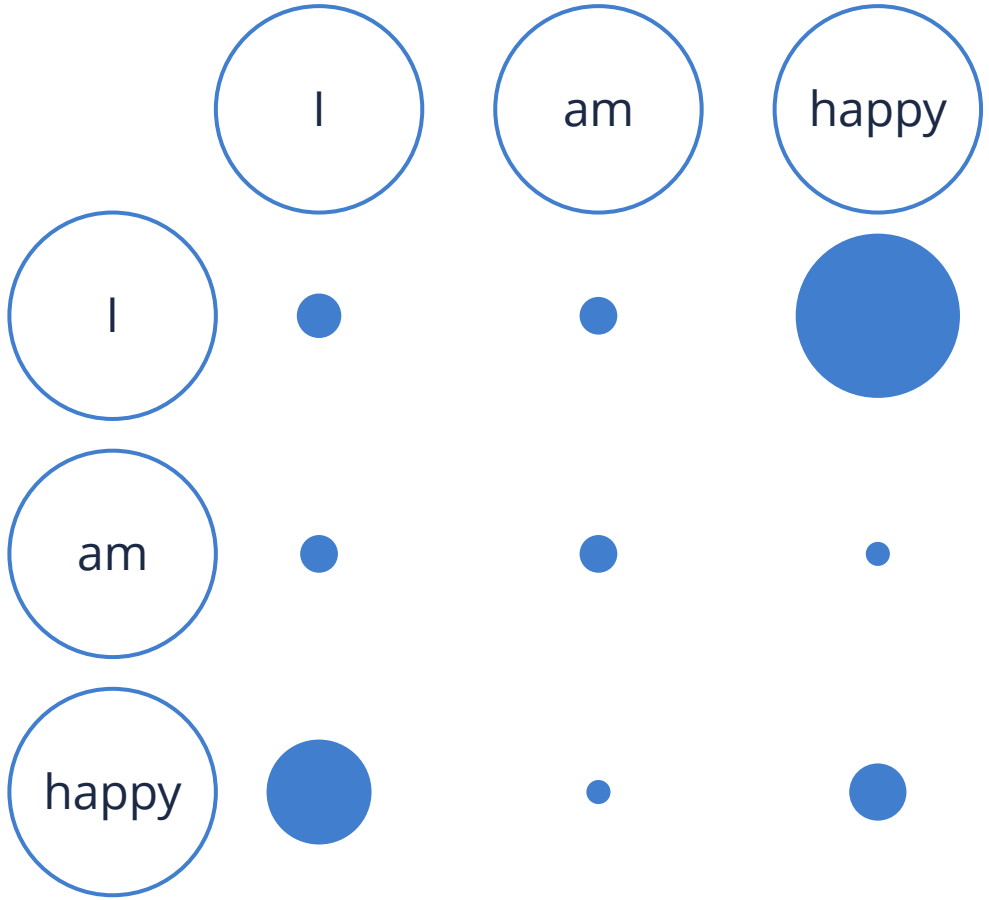


Transformers replace recurrence memory by attention

RNN
summary through time



Transformer
direct context lookup



Transformers are now widely used across biomedical data types

Article | Open access | Published: 28 January 2026

Advancing regulatory variant effect prediction with AlphaGenome

Žiga Avsec, Natasha Latysheva, Jun Cheng, Guido Novati, Kyle R. Taylor, Tom Ward, Clare Bycroft, Nicolaisen, Eirini Arvaniti, Joshua Pan, Raina Thomas, Vincent Dutordoir, Matteo Perino, Soham De, Alexander Karollus, Adam Gayoso, Toby Sargeant, Anne Mottram, Lai Hong Wong, Pavol Drotár, Ad Kosiorek, Andrew Senior, Richard Tanburn, Taylor Applebaum, ... Pushmeet Kohli + Show autho

Nature **649**, 1206–

284k Accesses | 1

Abstract

Deep learning m
are powerful too
trade-off betwee
modality scope a

Article | Open access | Published: 30 July 2024

Sequence-to-sequence translation from m to peptides with a transformer model

Melih Yilmaz, William E. Fondrie, Wout Bittremieux, Carlo F. Melendez, Rowan Ne Sewoong Oh & William Stafford Noble

Nature Communications **15**, Article number: 6427 (2024) | Cite this article

21k Accesses | **86** Citations | **66** Altmetric | Metrics

Abstract

A fundamental challenge in mass spectrometry-based proteomics is the identification of the peptide that generated each acquired tandem mass spectrum. Approaches that leverage known peptide sequence databases cannot detect unexpected peptides and can be impractical or impossible to apply in some settings. Thus, the ability to assign peptide sequences to tandem mass spectra without prior information—de novo peptide sequencing—is valuable for tasks including antibody sequencing, immunopeptidomics, and metaproteomics. Although many methods have been developed to address this problem, it

arXiv > cs > arXiv:2302.01588v1

Search...

Help | A

Computer Science > Computation and Language

[Submitted on 3 Feb 2023]

Bioformer: an efficient transformer language model for biomedical text mining

Li Fang, Qingyu Chen, Chih-Hsuan

Pretrained language models such as Bi
Transformers (BERT) have achieved st
processing (NLP) tasks. Recently, BER
Despite the effectiveness, these models
are computationally expensive when ap
hypothesized that the number of param
reduced with minor impact on performa

Article | Open access | Published: 08 May 2024

Accurate structure prediction of biomolecular interactions with AlphaFold 3

Josh Abramson, Jonas Adler, Jack Dunger, Richard Evans, Tim Green, Alexander Pritzel, Olaf Ronneberger, Lindsay Willmore, Andrew J. Ballard, Joshua Bambrick, Sebastian W. Bodenstein, David A. Evans, Chia-Chun Hung, Michael O'Neill, David Reiman, Kathryn Tunyasuvunakool, Zachary Wu, Akvilė Žemgulytė, Eirini Arvaniti, Charles Beattie, Ottavia Bertolli, Alex Bridgland, Alexey Cherepanov, Miles Congreve, ... John M. Jumper + Show authors

Nature **630**, 493–500 (2024) | Cite this article

1.29m Accesses | **11k** Citations | **2277** Altmetric | Metrics

An Addendum to this article was published on 27 November 2024

Abstract

The introduction of AlphaFold 2[1] has spurred a revolution in modelling the structure of

Recap

- Sequence data are ordered, so position carries meaning.
- Symbols must be converted into numbers, often through learned embeddings.
- RNNs process sequences step by step and maintain a hidden state as memory.
- Vanilla RNNs struggle with long-range dependencies; LSTMs and GRUs improve this.
- Transformers use attention to relate distant sequence elements directly.